



SACS

Interactive Fatigue

Version 24.00

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1 INTRODUCTION

1.1 GENERAL INFORMATION

The Interactive Fatigue program allows for reviewing of tubular connection fatigue results and redesigning connections in a 3D interactive environment. The program requires a fatigue extract file generated by the Fatigue batch program module as input. Various reports and plots of fatigue results can be generated and saved.

1.1.1 Program Features

Some of the main features of the Interactive Fatigue program are:

1. Shows the 3-D view of the connection and allows for braces to be selected with the mouse.
2. Allows for redesigning of chord and brace members based on fatigue results.
3. Allows SCF theory to be changed for any type connection, including in-line connections and connections with user defined SCFs.
4. Provides the ability to designate the S-N curve used to determine allowable cycles.
5. Recognizes all SCF and S-N options available in the batch program.
6. Full reporting capabilities.
7. Full plotting capabilities, including transfer function, response function, wave spectrum, stress range versus cycles and various other plots.
8. Reports and plots can be displayed on the screen or saved to a file.

1.2 PROGRAM OVERVIEW

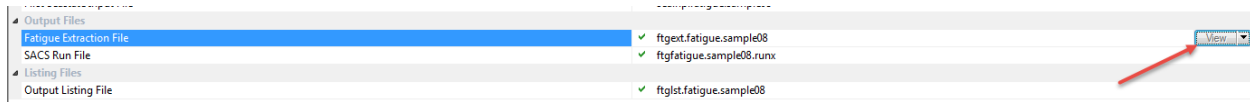
1.2.1 File Naming Conventions

While no file naming convention is required for Interactive Fatigue, the file name filter is used in the Open dialog box depending on the naming convention set in SACS Executive. The default filter is *???ext.** for the naming convention *pgmtyp.ver* and **.ext* for *verpgm.typ*. Plot filenames may be specified by the user using Save As under the File menu, and reports may be saved from the SACS report viewer program which is automatically called when reports are requested from the Reports menu.

2 GETTING STARTED

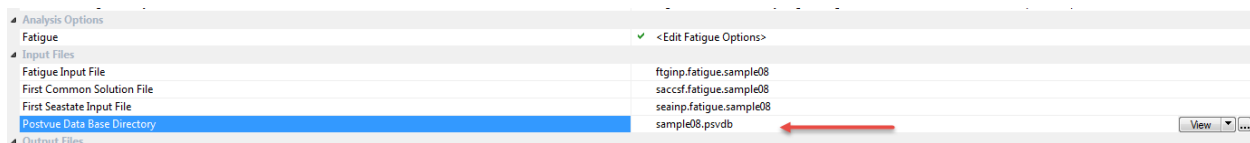
2.1 BEGINNING AN INTERACTIVE FATIGUE SESSION

To start the Interactive Fatigue program, go to Analysis Generator on Executive page and click the View drop down button and select Open with Interactive Fatigue, as shown below.

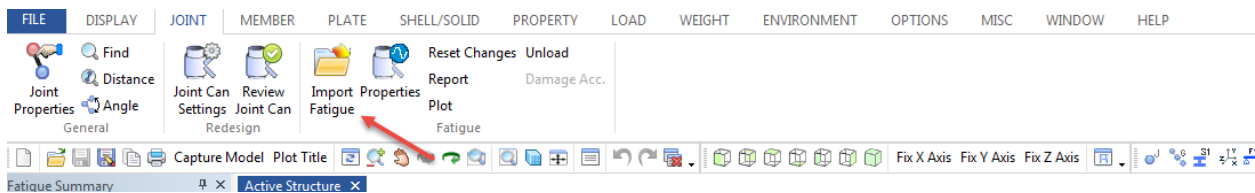


Alternatively, Fatigue Extraction can be dragged and dropped on Precede Modeler on the Executive page.

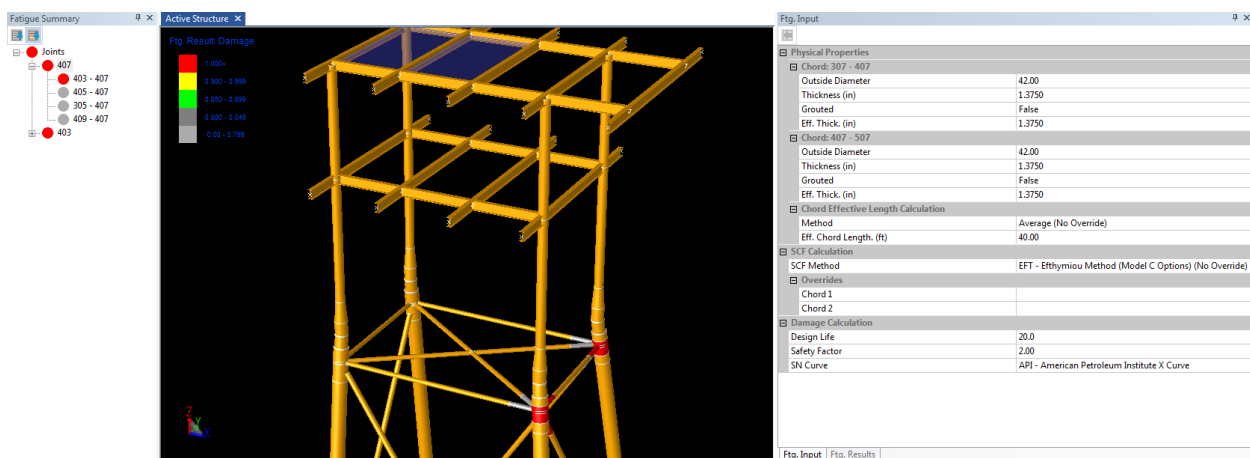
If fatigue results are to be displayed on the entire model, a Postvue database needs to be generated prior to fatigue analysis. Then, in column 20-21 of the FTOPT2 line enter PV. Load/reload the fatigue input file, there will be Postvue Database Directory where the file to be uploaded.



Upon the fatigue analysis completion, launch the Postvue database by double clicking or drag it onto Precede. Then, go to JOINT Tap and select Import Fatigue and upload a Fatigue Extract file to the model.



Fatigue results will be displayed on the model and users can view the fatigue damage results as shown below. Users can view the fatigue input and results, and also select the joints from the tree diagram.



2.2 FATIGUE CONNECTION COORDINATE SYSTEM

For each SACS load case making up the fatigue environment, fatigue results are evaluated at eight points spaced at 45 degrees around the perimeter of the welded connection on both the chord and brace side, see Figure 1 below:

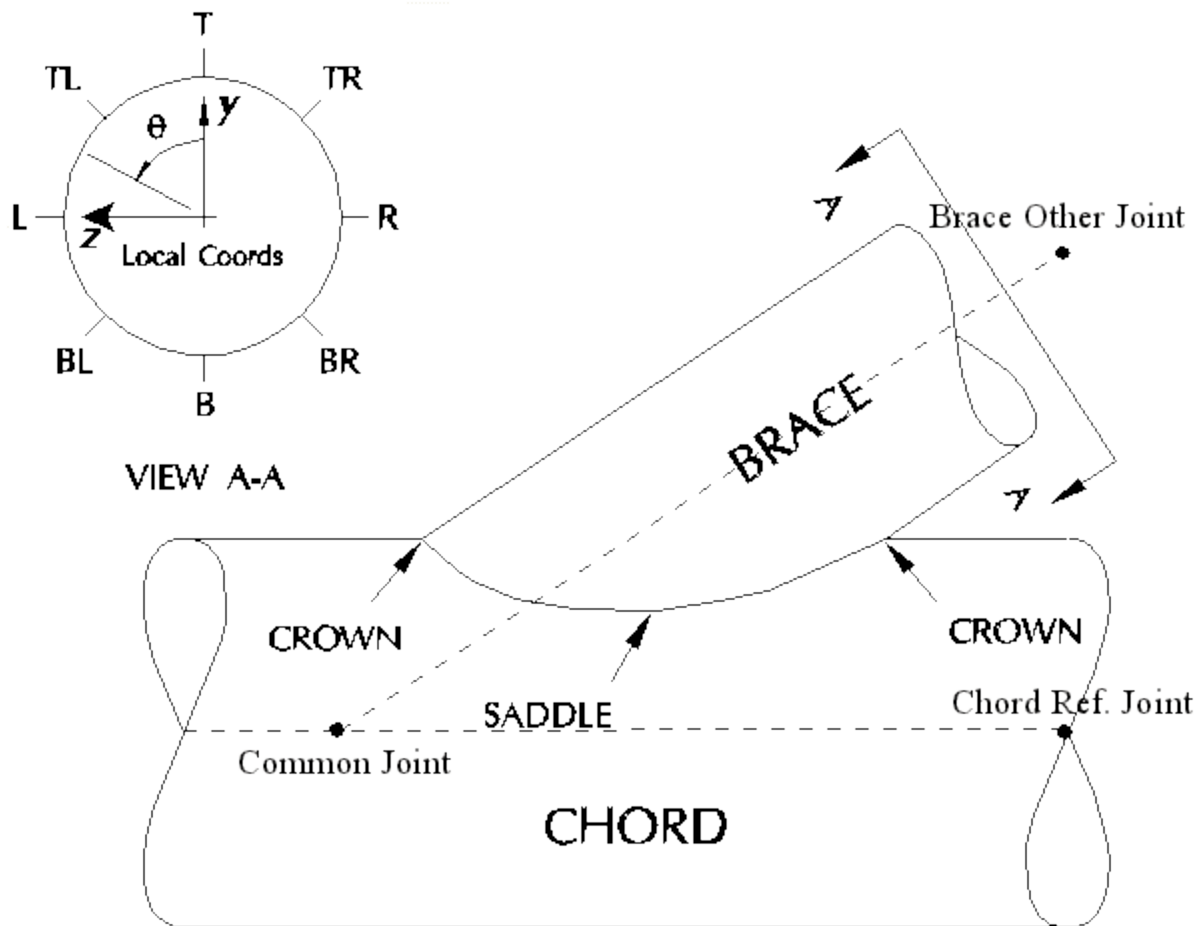


Figure 1

Note: The top of the connection is at the intersection of the brace/chord plane opposite of the chord reference joint. The Bottom position is therefore always at the intersection of the brace/chord plane on the side nearest to the chord reference joint.

2.3 EXITING THE INTERACTIVE FATIGUE SESSION

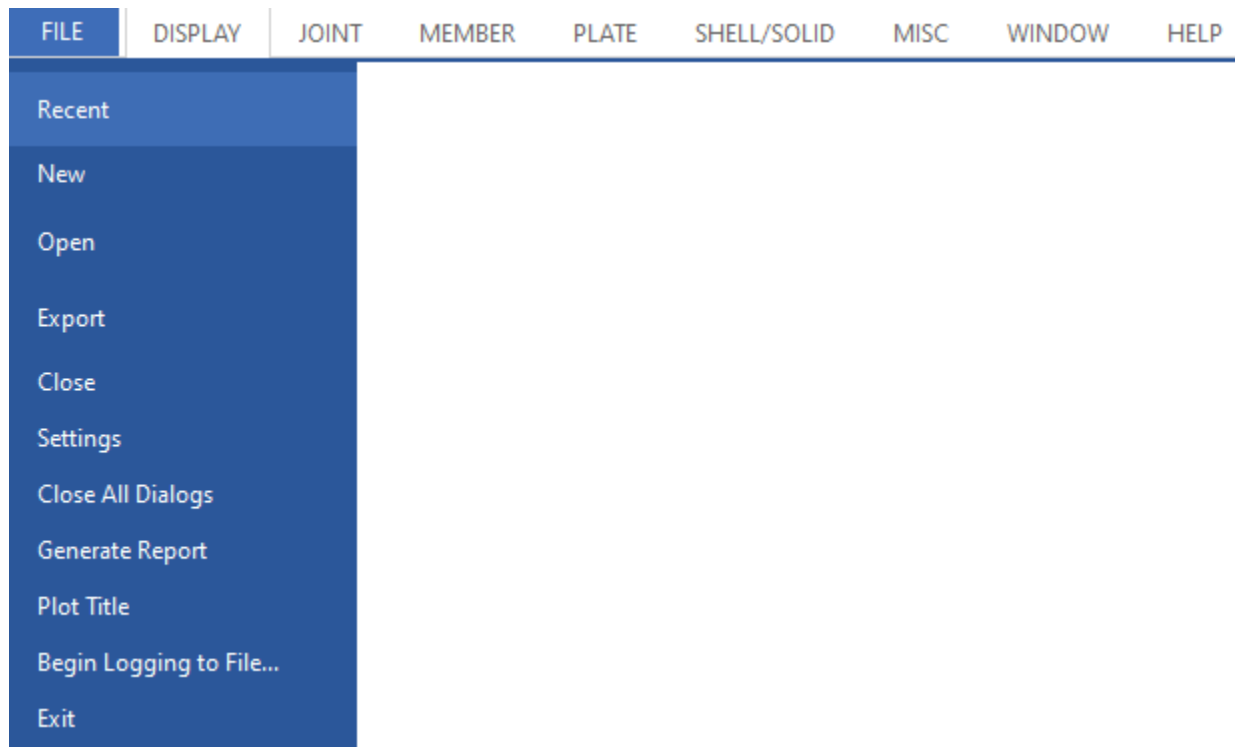
Select the File/Exit menu option to exit the Interactive Fatigue session.

If the fatigue extract has been imported within a Postvue database session, the fatigue data may be unloaded, leaving the Postvue session running.



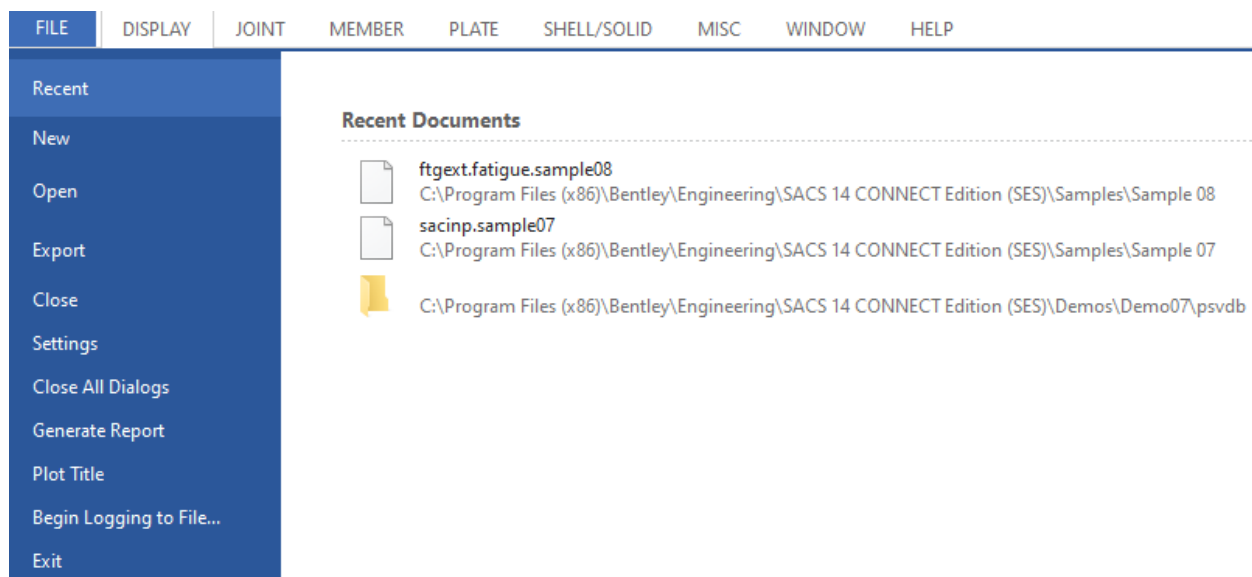
3 FILE MENU FEATURES

The following section details the commands found under the File main menu.



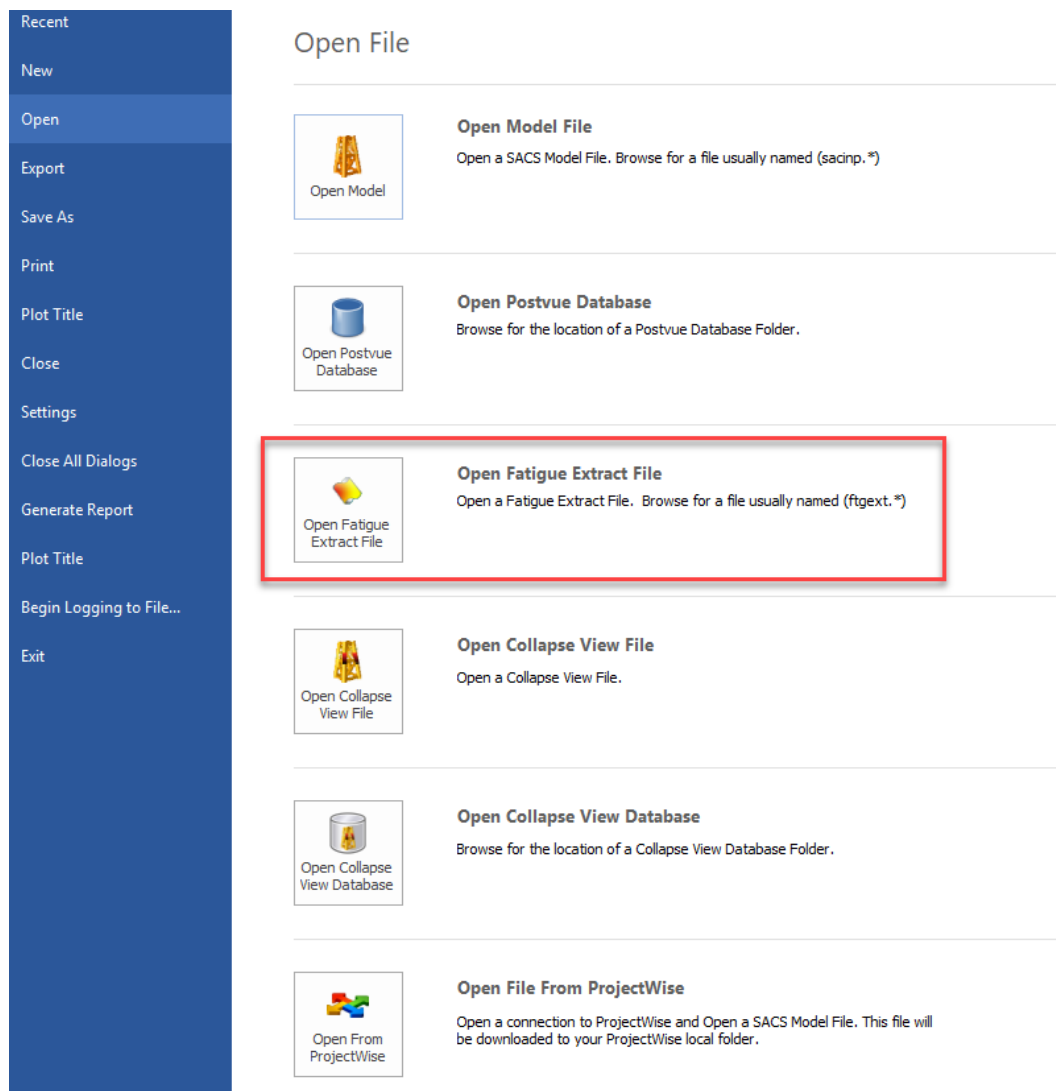
3.1 RECENT

The Recent option provides a quick access to recently used files list.



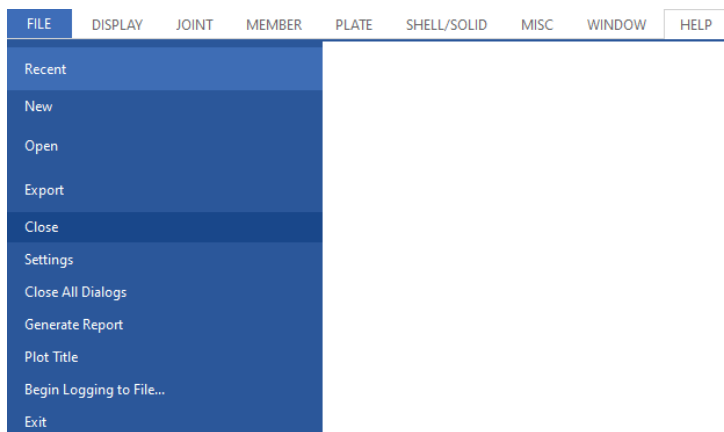
3.2 OPEN

Open is used to import an existing Fatigue extract file into Interactive Fatigue.



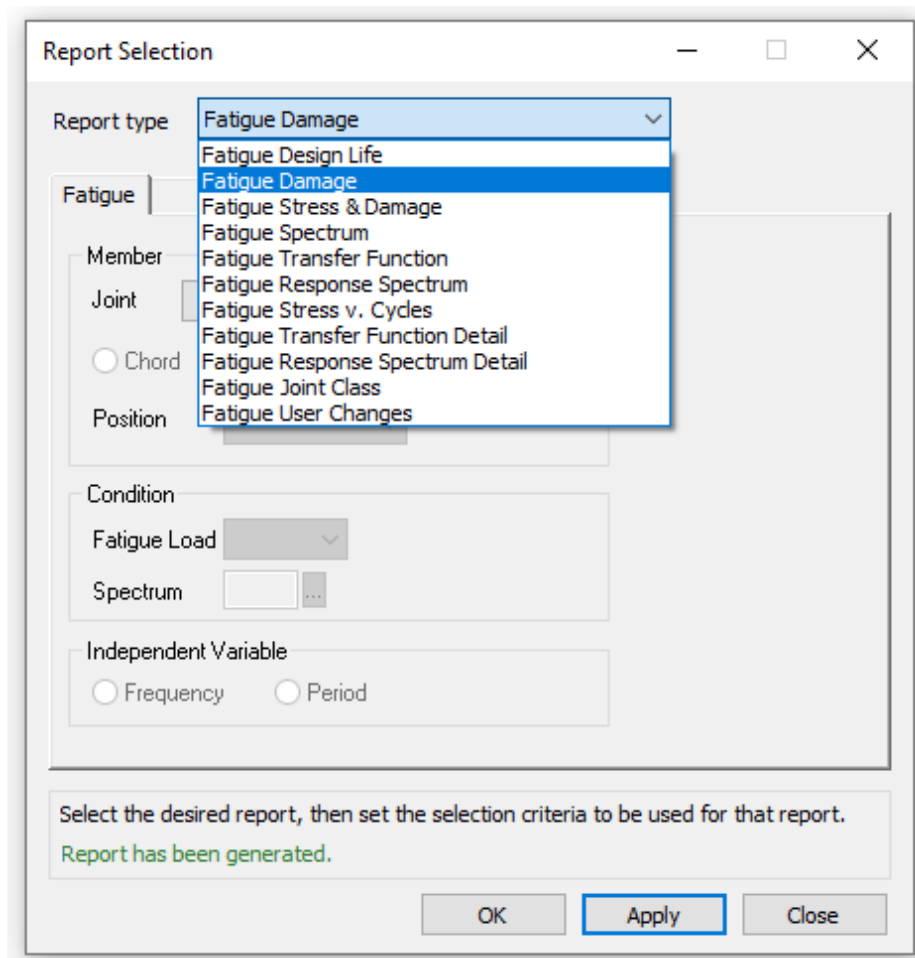
3.3 CLOSE

The Close option is used to close the current file.



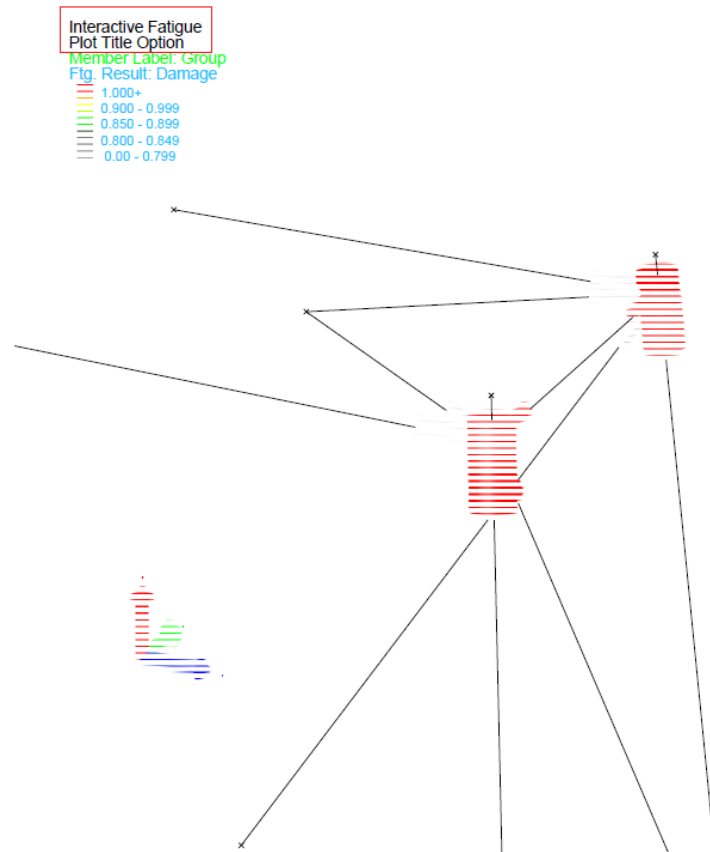
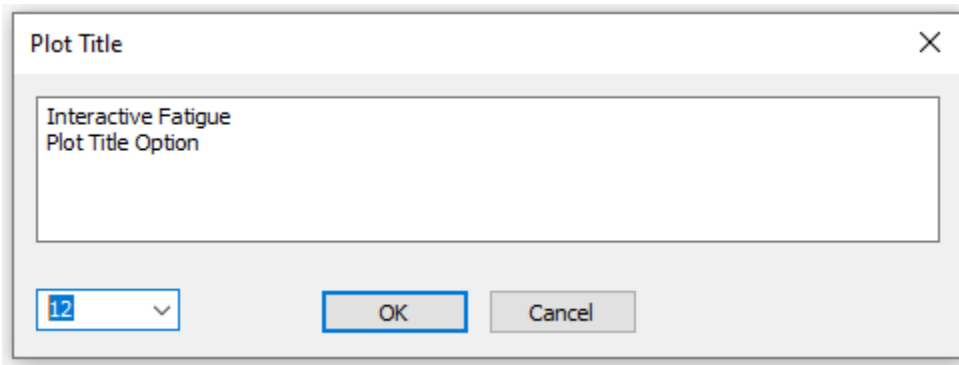
3.4 GENERATE REPORT

Generate Report option is used to generate and display a report in an output window.



3.5 PLOT TITLE

The Plot Title option is used to add a title to the current model view plots.

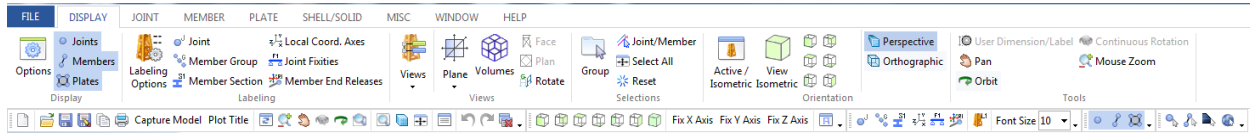


3.6 EXIT

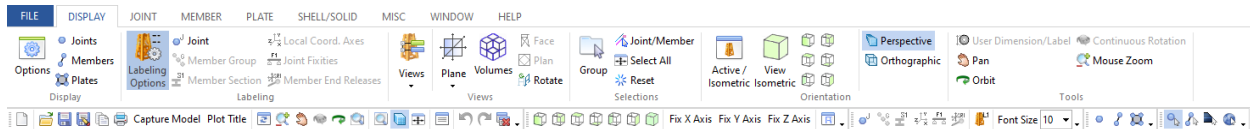
The Exit command terminates the interactive Fatigue session.

4 DISPLAY MENU FEATURES

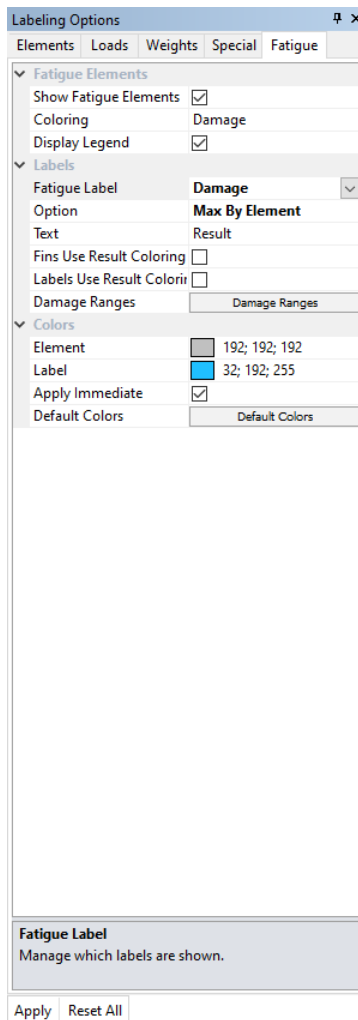
The Display menu features allow the user to specify or select display options including joint 3D view, joint plane view, rotation, fill, labeling, etc.



4.1.1 LABELING OPTIONS



The Labeling Options feature allows the user to display fatigue elements, labels and colors under the Fatigue tab as shown in the dialog below.



4.1.1.1 Show Fatigue Elements.

This option determines whether designed fatigue elements are included in the display.

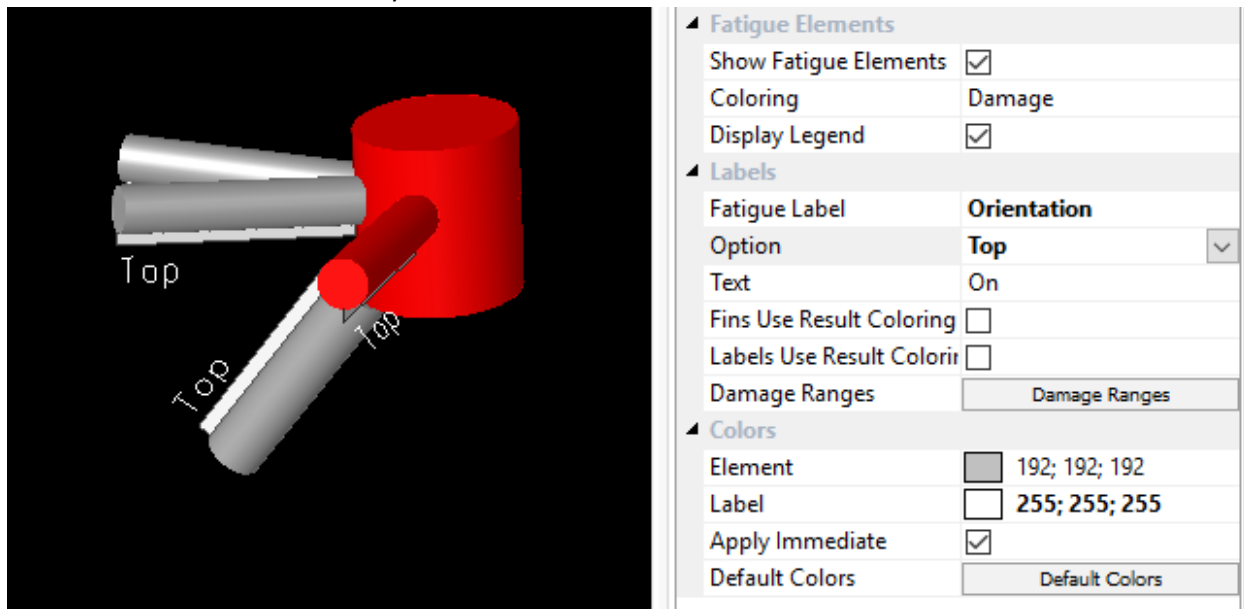
4.1.1.2 Coloring.

There are currently two options for element coloring. If Element is selected, all designed fatigue elements are shown in the element color. If Damage is selected, the elements will be color coded by damage level. In a chord-brace connection the chords are colored according to the worst brace result.

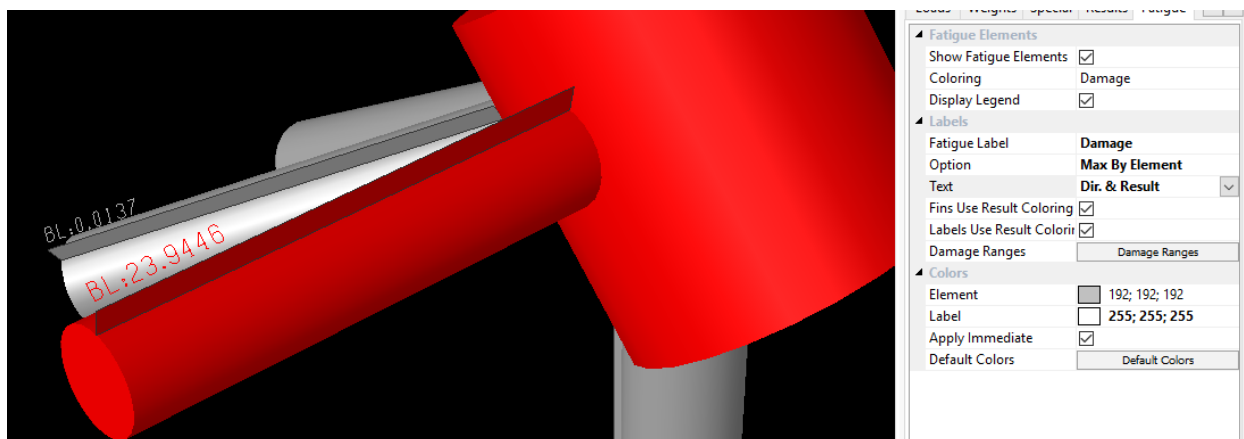
4.1.1.3 Fatigue Labeling.

A variety of labelling operations are available. Element orientation or damage labels may be selected.

Brace orientation is indicated by labelled fins.



Critical result per element.



4.1.1.3.1 Fatigue Label

This determines the general type of label, Damage, Orientation or None.

4.1.1.3.2 Option

This allows a number of type specific options. Orientation can denote Top or Top & Left directions. Damage allows a filtering of labels displayed. Max by Element will show one result on every designed element. Max By Joint shows the most critical result across all elements at each joint. > Top Band will label all results which exceed the top damage band (see below). All will label all elements at all eight design locations.

4.1.1.3.3 Text

This enables or disables the text part of the label. If set to Off, the fins will be drawn unlabeled. When displaying damage results, the fins may have the design location reported along with the damage.

4.1.1.3.4 Fins Use Result Coloring

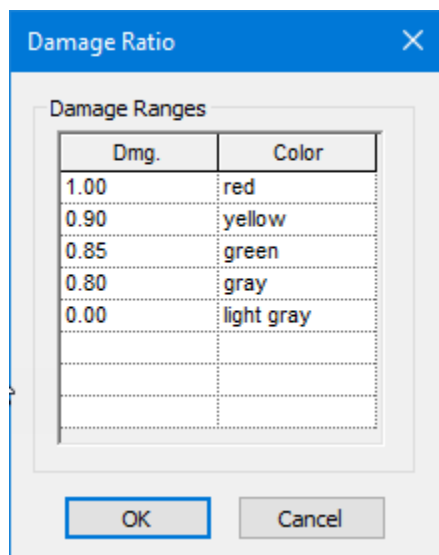
If checked, the fins will be colored according to the design result at that location. If unchecked they will be drawn in Label color.

4.1.1.3.5 Labels Use Result Coloring

If checked, the labels will be colored according to the design result at that location. If unchecked they will be drawn in Label color.

4.1.1.3.6 Damage Ranges

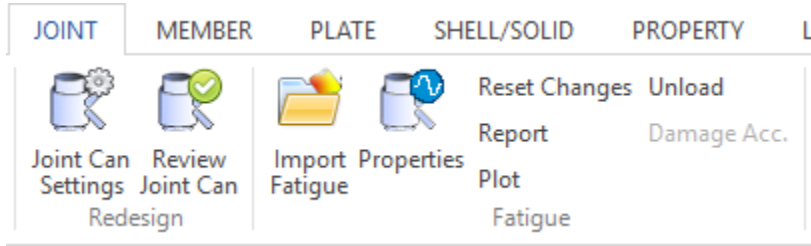
This allows the user to define the Damage:Color bands use when coloring the diagram



4.1.1.3.7 Colors

This section allows the user to set the colors used for elements and labels. Default Colors sets the element color to pale gray and the label color to a light blue

5 JOINT COMMANDS



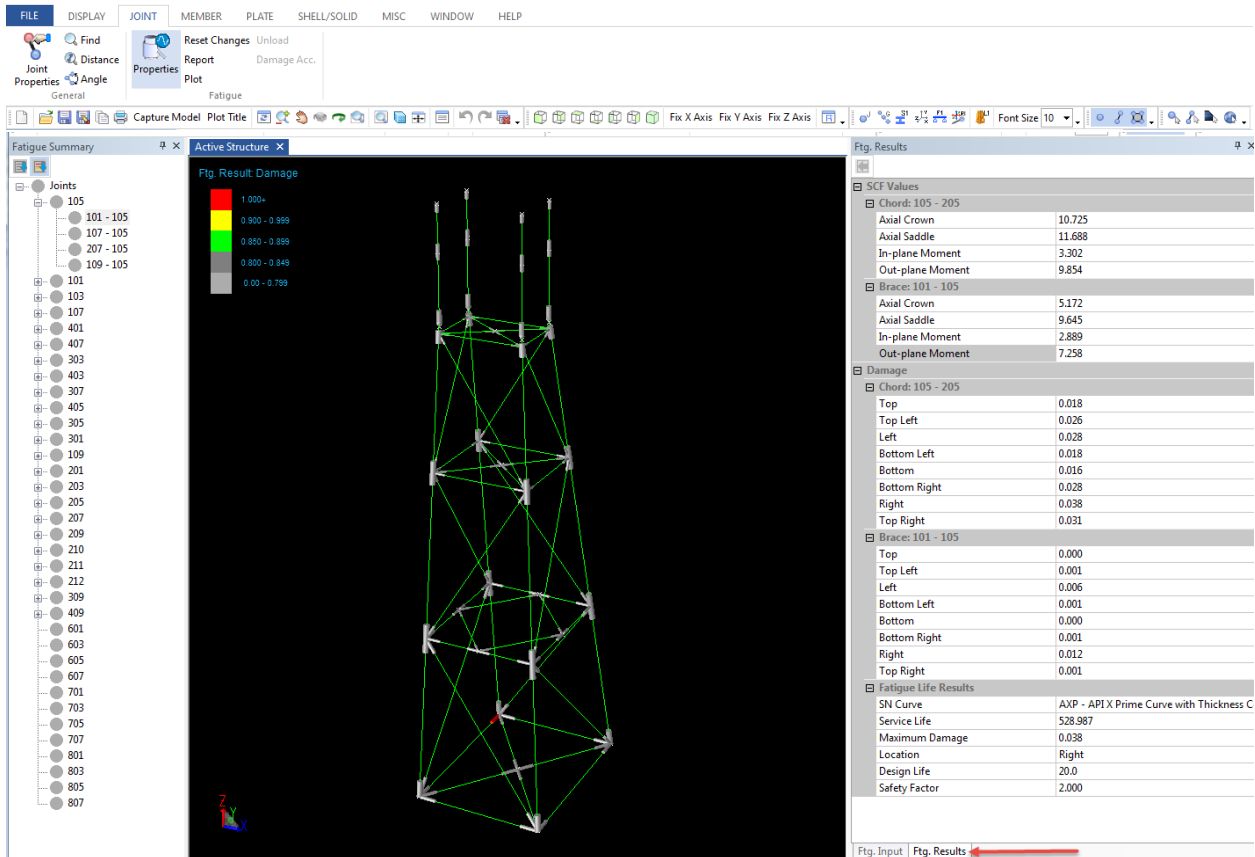
5.1 IMPORT FATIGUE

This allows a fatigue extract to be imported. If there is an extract already loaded, that will be unloaded before the new is loaded.

5.2 PROPERTIES

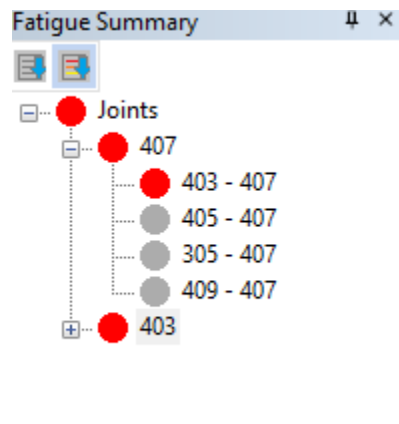
Onscreen textural data is divided over three dockable windows. The default arrangement is for the Fatigue Summary tree view to be on the left and the Fatigue Input and Fatigue Results panels to be stacked in a tabbed window on the right.

These windows are opened when a fatigue extract is first loaded. If they become closed, they can be reopened with the Properties command.



5.2.1 Fatigue Summary Tree

This panel shows a summary of the joints and members which are contained in the fatigue extract file.

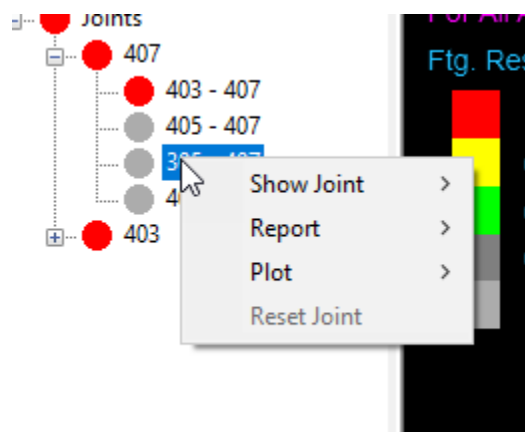


Each item is colored according to the current Damage:Color bands used by the graphical display. A higher level item is colored according to the worse case child item.

The two icons at the top of the view sort the joints by file read order (left) or by max damage (right). Braces, or member transition points are always under, and sorted within, their appropriate joint or member.

Selecting an item in this panel will cause the same item to be selected in the graphical view, and in the fatigue input & results panels. A selection of a joint will cause the chord element(s) to be drawn selected, but not the individual braces.

Joint, brace and transition point items in the tree support pop up menus. These are accessed with a mouse right-click.



The menu options are dependent on the level of the item being selected. Show Joint will create a new graphical view with only the selected joint visible. Report & Plot allow access to reports and charts for the selected item. Reset Joint will discard any changes made to the Joint's design parameters during this session.

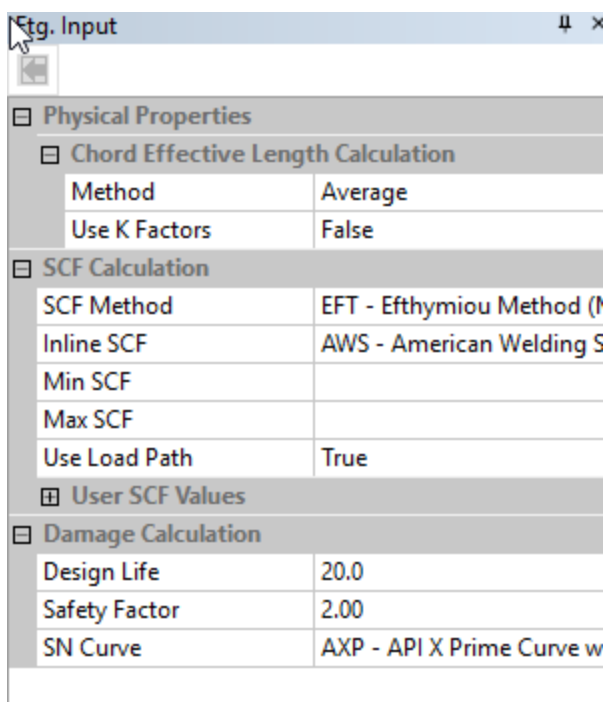
5.2.2 Fatigue Input Panel

This panel has a variety of forms dependent on the most recently selected fatigue item, either from the Summary tree panel or from the graphical display.

These property panels may be edited. When editing data, all relevant redesign is performed and the summary tree, graphical display and result panel are updated. Data which has changed by the user is shown in bold.

If the Joints or Members tree panel item is selected, the panel will show the general fatigue design parameters. Data here affects all joints and members which do not have specific overrides.

The icon at the top of the panel will reset the data to that first read from fatigue extract file.



The screenshot shows the 'Fatg. Input' panel with a reset icon (a square with a diagonal line) at the top left. The panel is organized into several sections, each with a collapse/expand icon (a square with a plus/minus sign) on the left. The sections are:

- Physical Properties**
 - Chord Effective Length Calculation**

Method	Average
Use K Factors	False
- SCF Calculation**

SCF Method	EFT - Efthymiou Method (I
Inline SCF	AWS - American Welding S
Min SCF	
Max SCF	
Use Load Path	True
- User SCF Values**
- Damage Calculation**

Design Life	20.0
Safety Factor	2.00
SN Curve	AXP - API X Prime Curve w

If a joint is selected in the fatigue summary tree or a chord selected in the graphical display, the input panel will show the design parameters relevant to that joint's design.

Physical Properties	
Chord: 307 - 407	
Outside Diameter	42.00
Thickness (in)	1.3750
Grouted	False
Eff. Thick. (in)	1.3750
Chord: 407 - 507	
Outside Diameter	42.00
Thickness (in)	1.3750
Grouted	False
Eff. Thick. (in)	1.3750
Chord Effective Length Calculation	
Method	Average (No Override)
Eff. Chord Length. (ft)	40.00
SCF Calculation	
SCF Method	EFT - Efthymiou Method (M
Overrides	
Chord 1	
Chord 2	
Damage Calculation	
Design Life	20.0
Safety Factor	2.00
SN Curve	API - American Petroleum In

Chord properties will affect the design of all brace elements at this joint. Data items with a text entry will indicate that they are inherited from the general options with the suffix (No Override).

If a brace is selected in the fatigue summary tree or in the graphical display, the input panel will show the design parameters relevant to that brace's design.

Ftg. Input

Physical Properties	
Chord: 407 - 507	
Outside Diameter	42.00
Thickness (in)	1.3750
Grouted	False
Eff. Thick. (in)	1.3750
Brace: 405 - 407	
Outside Diameter	12.75
Thickness (in)	0.5000
Grouted	False
Eff. Thick. (in)	0.5000
Connection	
Type	K
Brace Angle	84.33
Gap (in)	9.383
Eff. Chord Length. (ft)	40.00
SCF Calculation	
Overrides	
Brace	
Chord	
Connection Overrides	
Brace: Axial Crown	
Brace: Axial Saddle	
Brace: In-plane Momen	
Brace: Out-plane Momen	
Chord: Axial Crown	
Chord: Axial Saddle	
Chord: In-plane Momen	
Chord: Out-plane Momen	
Damage Calculation	
Design Life	20.0
Safety Factor	2.00
SN Curve	AXP - API X Prime Curve with

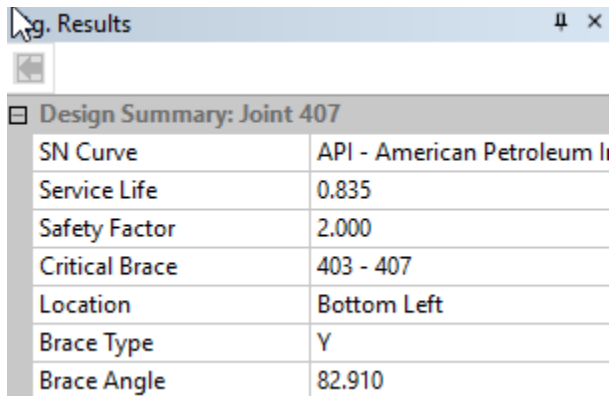
This follows the principals above. Changing the chord properties will alter that chord element's properties when other braces at that joint are redesigned. If using the LDPATH option the connection type will not be editable.

There are similar input property panels for inline joint design and for member transition point design.

5.2.3 Fatigue Result Panel

When the selection in the summary tree is a joint or if the graphical selection is a chord element, this will show summary result information for the joint.

If a joint is selected in the fatigue summary tree or a chord selected in the graphical display, the input panel will show the design parameters relevant to that joint's design.

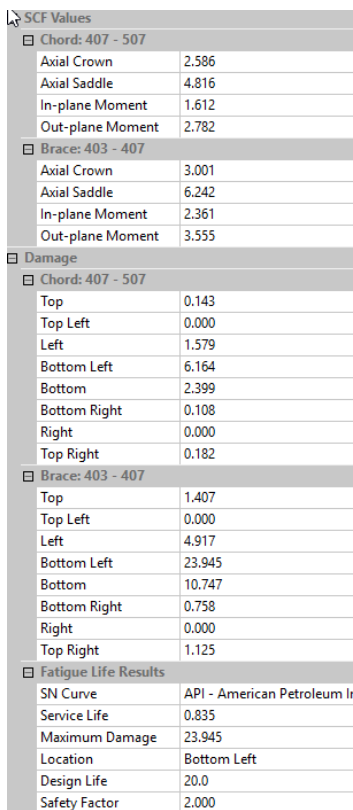


The screenshot shows a window titled 'g. Results' with a close button. Below the title bar is a 'Design Summary: Joint 407' table with the following data:

SN Curve	API - American Petroleum I
Service Life	0.835
Safety Factor	2.000
Critical Brace	403 - 407
Location	Bottom Left
Brace Type	Y
Brace Angle	82.910

When the selection in the summary tree is a brace or if the graphical selection is a brace element, this will show result information for the joint.

When the selection in the summary tree is a joint or if the graphical selection is a chord element, this will show summary result information for the joint.



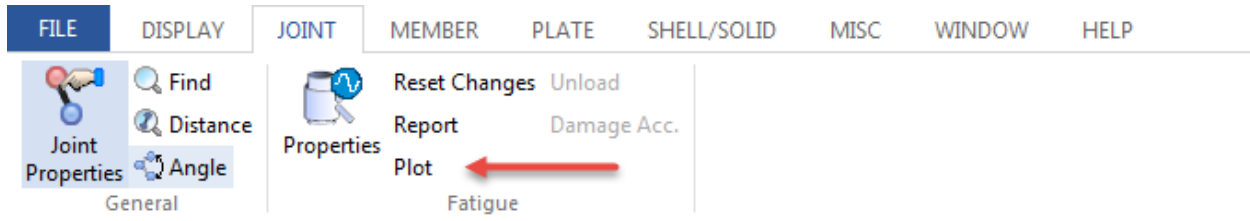
The screenshot shows a window titled 'SCF Values' with a tree view. The selected item is 'Chord: 407 - 507'. The table below shows the following data:

Chord: 407 - 507	
Axial Crown	2.586
Axial Saddle	4.816
In-plane Moment	1.612
Out-plane Moment	2.782
Brace: 403 - 407	
Axial Crown	3.001
Axial Saddle	6.242
In-plane Moment	2.361
Out-plane Moment	3.555
Damage	
Chord: 407 - 507	
Top	0.143
Top Left	0.000
Left	1.579
Bottom Left	6.164
Bottom	2.399
Bottom Right	0.108
Right	0.000
Top Right	0.182
Brace: 403 - 407	
Top	1.407
Top Left	0.000
Left	4.917
Bottom Left	23.945
Bottom	10.747
Bottom Right	0.758
Right	0.000
Top Right	1.125
Fatigue Life Results	
SN Curve	API - American Petroleum I
Service Life	0.835
Maximum Damage	23.945
Location	Bottom Left
Design Life	20.0
Safety Factor	2.000

There are similar result property panels for inline joint design and for member transition point design.

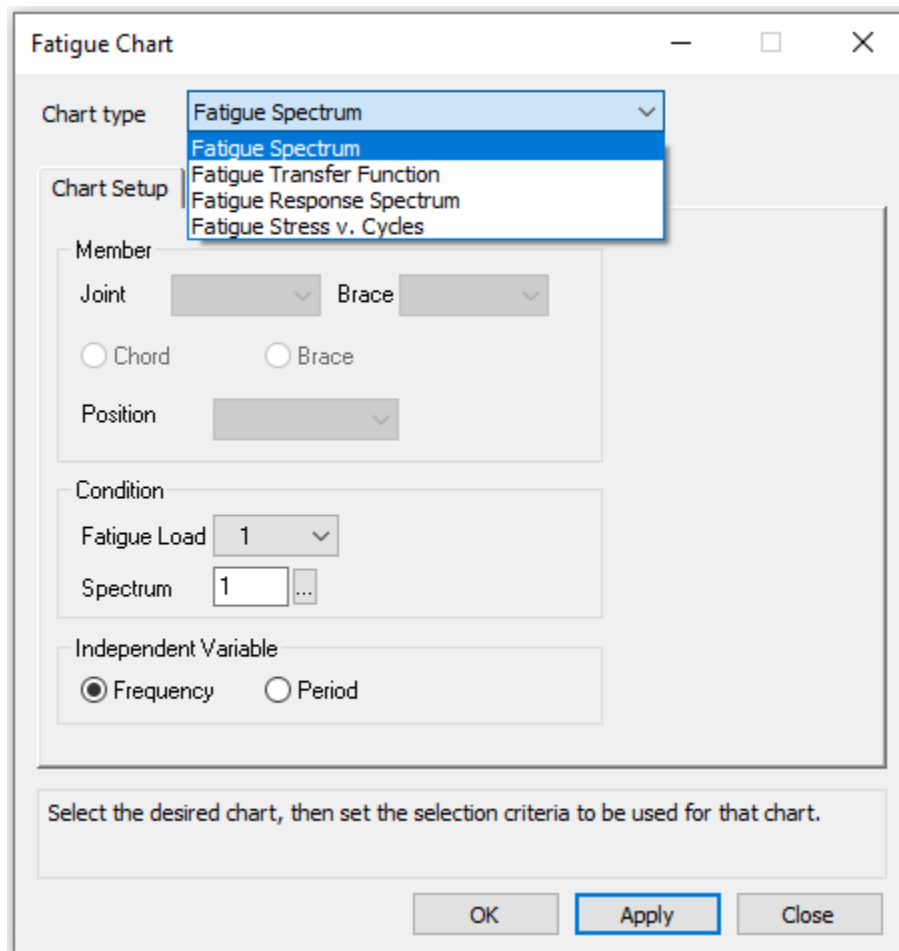
5.3 PLOT

The Plot menu features allow the user to generate various plots and graphs for display on the screen or to save in a plot file. A plot file *iftgncf* will be created to store all exported plots.



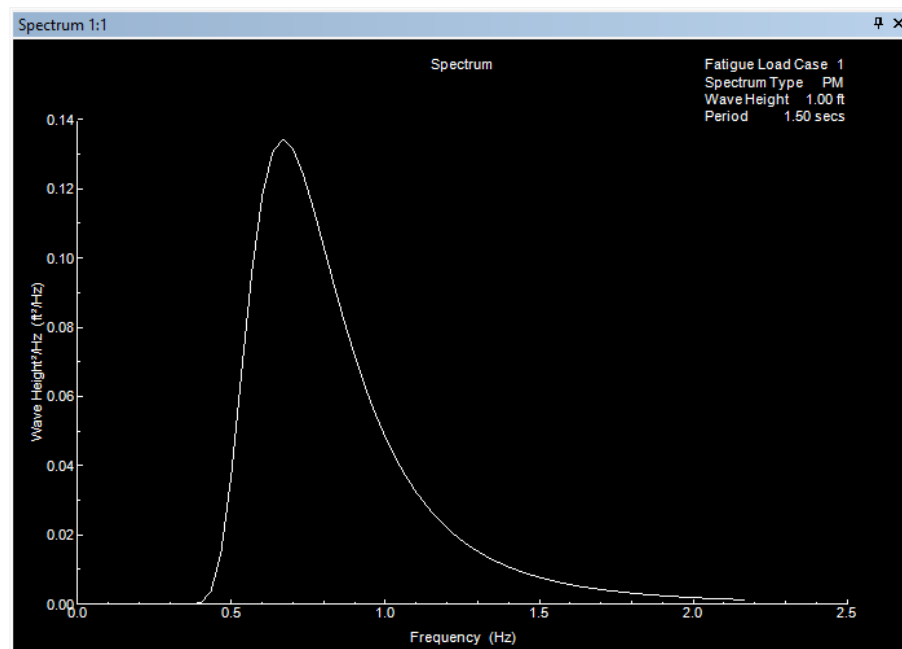
Select the desired plot options including the element to be plotted, i.e. chord or brace, and position around the circumference. The desired Fatigue load case and if applicable the desired wave spectrum may be designated.

Note: Plots are generated and displayed on the screen by default. They may be added to the chartvue file (iftgncf) from a right-click menu on the screen chart.



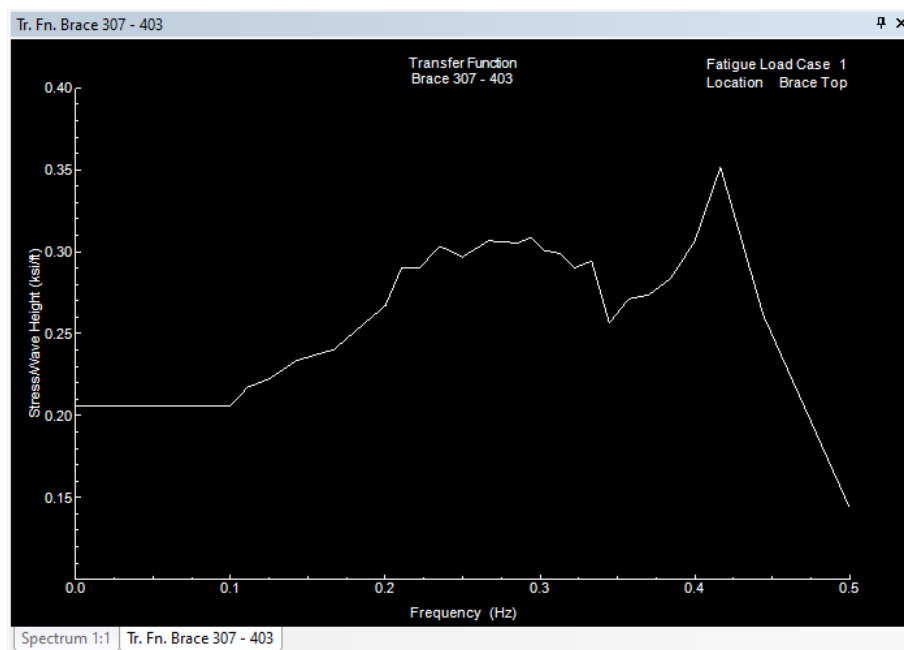
5.3.1 FATIGUE SPECTRUM

The Spectrum plot command plots the wave spectrum designated on the Fatigue Chart dialog. The Member and Position plot options are not applicable to this feature.



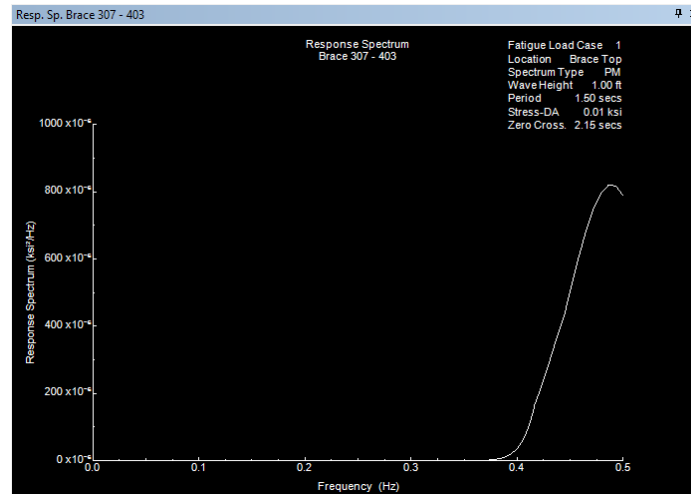
5.3.2 FATIGUE TRANSFER FUNCTION

Select the Fatigue Transfer Function command to plot the cyclic hot-spot stress transfer function of the chord or brace position designated and for the particular Fatigue Load Case (wave direction) specified on the Plot Options dialog.



5.3.3 FATIGUE RESPONSE SPECTRUM

Select the Response Spectrum command to plot the response spectrum of the chord or brace position designated. The applicable Fatigue Load Case and Wave spectrum are designated on the Plot Options dialog.

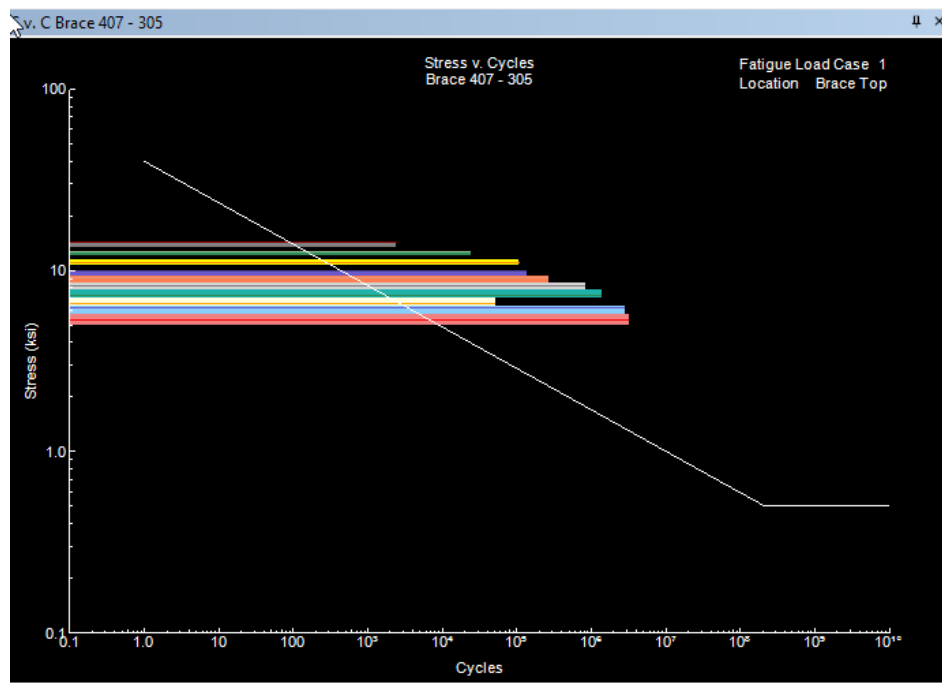


5.3.4 FATIGUE STRESS V. CYCLES

The Stress vs. Cycles command creates a bar graph showing stress range and number of cycles plotted against the S-N allowable curve. The chord or brace position along with the applicable fatigue load case must be designated on the Plot Options dialog.

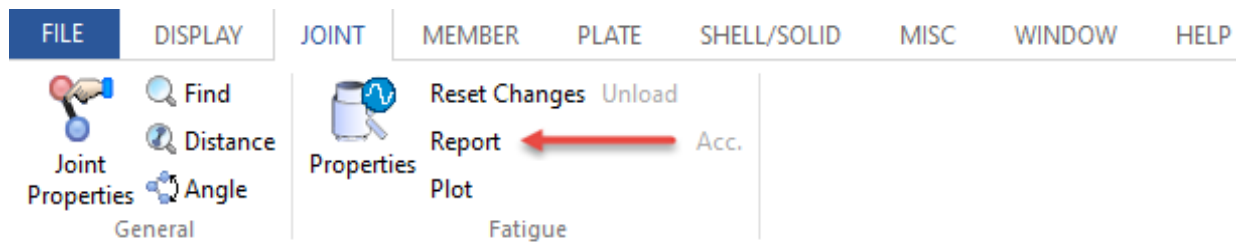
The Stress vs. Cycles plot is generated as follows:

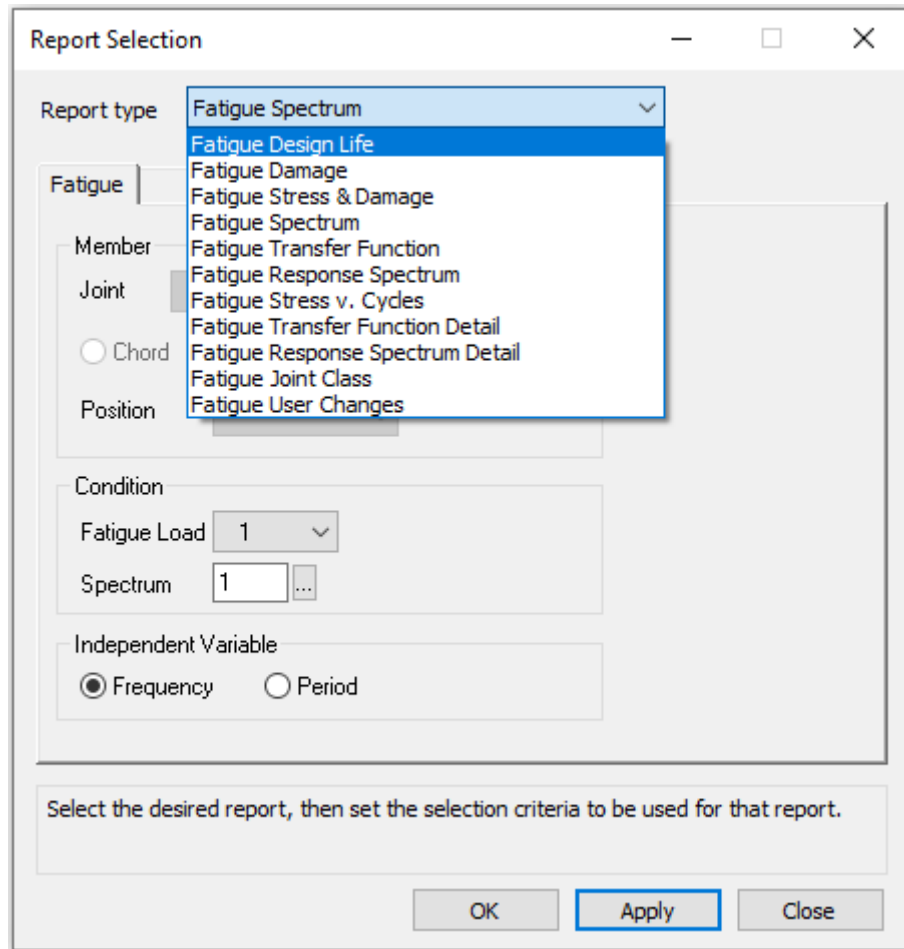
For a point on the connection, the maximum cyclic stress is found considering all spectra for the direction in question. This maximum stress is then broken into a number of cyclic stress bands or categories. For each band, the total damage and number of cycles is summed. The report and plot shows the range and cycles for a band only if the band causes actual damage .



5.4 REPORT

The Report menu features allow the user to generate various reports.





The image shows a 'Report Selection' dialog box with a title bar containing standard window controls. The 'Report type' dropdown menu is open, displaying a list of report options: 'Fatigue Spectrum' (selected), 'Fatigue Design Life', 'Fatigue Damage', 'Fatigue Stress & Damage', 'Fatigue Spectrum', 'Fatigue Transfer Function', 'Fatigue Response Spectrum', 'Fatigue Stress v. Cycles', 'Fatigue Transfer Function Detail', 'Fatigue Response Spectrum Detail', 'Fatigue Joint Class', and 'Fatigue User Changes'. On the left side of the dialog, there are several sections: 'Fatigue' (a tab), 'Member' (a dropdown menu), 'Joint' (a dropdown menu), 'Chord' (a radio button), and 'Position' (a dropdown menu). Below these, the 'Condition' section contains 'Fatigue Load' (a dropdown menu with '1' selected) and 'Spectrum' (a dropdown menu with '1' selected and a '...' button). The 'Independent Variable' section has two radio buttons: 'Frequency' (selected) and 'Period'. At the bottom of the dialog, there is a text box with the instruction 'Select the desired report, then set the selection criteria to be used for that report.' and three buttons: 'OK', 'Apply' (highlighted with a blue border), and 'Close'.

Reports may be displayed on the screen in the Report Browser and from there may be saved to a file.

5.4.1 FATIGUE DESIGN LIFE

The Fatigue Design Life command creates a detailed report for the active joint and brace. The report contains basic properties of the chord and brace, fatigue life and the damage reported at each of the eight positions. The following is a sample Fatigue Life report:

Fatigue Life results at Joint 101							
Brace 101- 203 Chord 101- 201							
Fatigue Life (years)	796.385	SCF Method	Efthymiou Method				
Design Life (years)	20.000	Life Safety Fact	2.000				
Maximum Damage	0.025	Location	Bottom				
Brace-Chord Angle (deg)	30.913	Connection Type	K	Gap (Cm)	14.003		
*** Brace and Chord Properties ***							
Member	Chord	Diameter	Thick	*** Stress Concentration Factors ***			
	Brace	(Cm)	(Cm)	Ax-Crn	Ax-Sdl	IP-Mom	OP-Mom
101- 201	Chord	105.000	3.500	4.204	2.919	1.151	1.495
101- 203	Brace	50.000	2.000	3.933	2.265	2.800	1.399
*** Damage and Fatigue Life Details ***							
Position	***** CHORD *****		***** BRACE *****				
	Damage	Life	Damage	Life			
Top	.023662	845.236	.013024	1535.67			
Top Left	.7728-2	2588.11	.2266-2	8827.50			
Left	.5764-3	34700.5	.1230-4	1626.+3			
Bottom Left	.7998-2	2500.55	.3062-2	6530.88			
Bottom	.025113	796.385	.019896	1005.23			
Bottom Right	.7791-2	2567.04	.2965-2	6745.00			
Right	.5730-3	34902.6	.1328-4	1506.+3			
Top Right	.7542-2	2651.88	.2652-2	7540.43			
Maximum	.025113	796.385	.019896	1005.23			

5.4.2 FATIGUE DAMAGE

The Damage report command generates a detailed report showing damage at each of the eight positions around the circumference of the connection caused by each wave spectrum.

Fatigue Damage Details at Joint 101										
Chord 101- 201			Brace 101- 203							
Ftg LC	Spect No.	Chord Brace	Top	Top Left	Left	Bottom Left	Bottom	Bottom Right	Right	Top Right
1	1	C	.0000000	.0000000	.0000000	.0000000	.0000000	.0000000	.0000000	.0000000
		B	.0000000	.0000000	.0000000	.0000000	.0000000	.0000000	.0000000	.0000000
1	2	C	.19072-4	.14068-5	.0000000	.11842-5	.15504-4	.12271-5	.0000000	.14561-5
		B	.17551-4	.0000000	.0000000	.0000000	.14194-5	.0000000	.0000000	.0000000
2	1	C	.0000000	.0000000	.0000000	.0000000	.0000000	.0000000	.0000000	.0000000
		B	.0000000	.0000000	.0000000	.0000000	.0000000	.0000000	.0000000	.0000000

5.4.3 FATIGUE STRESS & DAMAGE

The Stress and Damage reports shows RMS stress and Damage at each chord and brace position around the circumference of the connection for each of the wave spectra in the Fatigue Load case specified by the user.

Fatigue RMS Stress and Damage at Joint 101					
Chord 101- 201 Brace 101- 203					
Fatigue Load Case	Spectrum Number	Position	Chord Brace	RMS Stress (KG/CM2)	Damage
1	1	Bottom	Brace	0.234624	.14194-5
	1	Bottom Left	Brace	0.190174	.0000000
	1	Left	Brace	0.138432	.0000000
	1	Top Left	Brace	0.219104	.0000000
	1	Top	Brace	0.275573	.17551-4
	1	Top Right	Brace	0.221414	.0000000
	1	Right	Brace	0.141655	.0000000
	1	Bottom Right	Brace	0.192416	.0000000
	1	Bottom	Chord	0.263619	.15504-4
	1	Bottom Left	Chord	0.224893	.11842-5
	1	Left	Chord	0.181843	.0000000
	1	Top Left	Chord	0.236798	.14068-5
	1	Top	Chord	0.280472	.19072-4
	1	Top Right	Chord	0.239245	.14561-5
	1	Right	Chord	0.185285	.0000000
	1	Bottom Right	Chord	0.227313	.12271-5
1	2	Bottom	Brace	0.291327	.16509-2
	2	Bottom Left	Brace	0.237372	.32396-4
	2	Left	Brace	0.163807	.0000000
	2	Top Left	Brace	0.237601	.33213-4
	2	Top	Brace	0.291674	.16994-2
	2	Top Right	Brace	0.233524	.31164-4
	2	Right	Brace	0.158029	.0000000
	2	Bottom Right	Brace	0.233282	.30377-4
	2	Bottom	Chord	0.311029	.21247-2
	2	Bottom Left	Chord	0.267762	.36991-3
	2	Left	Chord	0.213636	.0000000
	2	Top Left	Chord	0.267857	.37329-3
	2	Top	Chord	0.311171	.21485-2
	2	Top Right	Chord	0.263496	.35137-3
	2	Right	Chord	0.207463	.0000000
	2	Bottom Right	Chord	0.263396	.34811-3

5.4.4 FATIGUE SPECTRUM

The Spectrum report shows the points of the wave spectrum selected by the user in tabular format.

5.4.5 FATIGUE TRANSFER FUNCTION

The Transfer Function report shows the points of the transfer function for the selected point on the chord or brace. The chord or brace position along with the Fatigue load case are specified on the Report Options dialog.

Note: Designation of the wave spectrum is not required for the transfer function report.

Transfer Function Report				
Fatigue Load Case 1 Chord 101- 201 Position Bottom				
Wave	Height (M)	H.S. Stress Range (KGSCM)	HSSR/Height (KGSCM *M)	Frequency (Hz)
-----	-----	-----	-----	-----
1	10.300	148.953	14.461	0.10000
2	6.620	111.227	16.802	0.12500
3	3.720	77.895	20.940	0.16667
4	2.590	34.808	13.440	0.20000
5	1.650	56.084	33.991	0.25000
6	0.930	103.111	110.872	0.33333
7	0.520	130.998	251.920	0.44444
8	0.410	57.396	139.992	0.50000
9	0.320	144.331	451.035	0.57143
10	0.230	54.496	236.941	0.66667

5.4.6 FATIGUE RESPONSE SPECTRUM

The Response Spectrum report shows the points of the response function for the selected point on the chord or brace for the specified wave spectrum. The chord or brace position along with the Fatigue load case and the appropriate wave spectrum are specified on the Report Options dialog.

5.4.7 FATIGUE STRESS VS CYCLES

Stress Range vs. Cycle Report			
Fatigue Load Case 1 Chord 101- 201 Position Bottom			
	Range Number	Stress Range (KGSCM)	Cycles
	-----	-----	-----
1515367	9	257.1	
	10	288.7	481573
	11	314.7	231189
	12	347.3	143603
	13	354.6	17075
	14	396.2	39501
	15	423.9	6830
	16	453.8	3096
	17	486.9	1536
	19	552.7	153

The Fatigue Stress vs. Cycles report contains hot-spot stress ranges and number of cycles for the selected point on the chord or brace. The chord or brace position along with the Fatigue load case are specified on the Report Options dialog.

The transfer function detail report contains axial load and bending moment normalized by wave height. The report also contains real and imaginary roots.

5.4.9 FATIGUE RESPONSE SPECTRUM DETAIL

Response Function Detail Report					
Member		101-	203	Fatigue Load Case	1 Spectrum 1
***** Axial Load *****		** Out-Plane Moment **		*** In-Plane Moment ***	
RMS	Zero Cross	RMS	Zero Cross	RMS	Zero Cross
(KG)	(sec)	(CM-KG)	(sec)	(CM-KG)	(sec)
-----	-----	-----	-----	-----	-----
466.213	1.714	400.529	451.840	1.695	1.665
FREQUENCY		MOMENT		MOMENT	
HZ	KG **2/HZ	OUT-PLANE	CM-KG **2/HZ	IN-PLANE	CM-KG **2/HZ
-----	-----	-----	-----	-----	-----
0.3611	65.110	2.728		76.051	
0.3750	440.079	23.890		457.900	
0.3889	2118.480	142.698		1965.001	
0.4028	7768.175	628.675		6426.693	
0.4167	22861.861	2168.366		16875.625	
0.4306	56261.289	6131.685		37060.078	
0.4444	119589.320	14739.152		70295.234	

5.4.10 FATIGUE JOINT CLASS

The Joint Class report shows the fraction of the joint types applicable when determining SCFs for each load case used to develop the transfer function.

Joint Classification Report for Joint 101				
LOAD CASE	***** TY	***** X	***** KT	***** K1
1	0.891	0.000	0.000	0.109
2	0.895	0.000	0.000	0.105
3	0.906	0.000	0.000	0.094
4	0.568	0.000	0.000	0.432
5	0.864	0.000	0.000	0.136
6	0.879	0.000	0.000	0.121
7	0.882	0.000	0.000	0.118
8	0.884	0.000	0.000	0.116

5.4.11 FATIGUE USER CHANGES

Fatigue User Changes report provides all the changes made in the Fatigue Input file.

Ftg. Input

Physical Properties

Chord Effective Length Calculation

MethodAverage

Use K FactorsFalse

SCF Calculation

SCF MethodEFT - Efthymiou Method (Model C Options)

Inline SCFAWS - American Welding Society

Min SCFAWS - American Welding Society

Max SCFDNV - Det Norske Veritas

Use Load PathDN2 - DnV-RP-C203 2005 formula

User SCF ValuesDN3 - DNV-RP-C203 2008 formula

Damage Calculation

Design LifeDE - Department of Energy

Safety FactorNS1 - Norsok Eq. C.2.7

SN CurveNS2 - Norsok Eq. C.2.10

MDX - Maddox

SHL - Shell Hague

Fatigue User Changes

General Parameters

Parameter	New Value	Original Value
SCF Calculation		
Inline Method	Det Norske Veritas	American Welding Society

5.5 RESET CHANGES

This reverses any user made changes to the fatigue input parameters and redesigns according to the data in the fatigue extract file.

5.6 UNLOAD

Removes the currently loaded fatigue extract information from the current PostVue session

5.7 DAMAGE ACC. (Accumulation)

This menu is only applicable when using a fatigue extract summation file. This file is auto generated by adding the extract line to the partial fatigue input file used in sacs wind turbine and GHBladed processes. Refer to Wind Turbine manual section 2 and 3 for details. User can create this file manually to sum up damages from different fatigue runs. It should be noted that in order to use this option all joint should be extracted by entering 'AL' in columns 14-15 of EXTRACT HEAD line.

The Fatigue Extract summation files as shown below contains the following information:

SUMFTGIFT: Flag indicating a summation extract file
NCASES: Defines the number of simulation for analysis cases.
NGRPS: Defines the number of groups for analysis cases.
DIRECTORY: Directory location for fatigue extract file for a fatigue simulation
EXTRCT: Name of the extract file generated from above run
GROUP: Defines the group above run belongs to
Repeat the above (Directory to Group) for other simulations (total equals NCASES)
JOINTS: Defines the number of joints to be viewed in interactive fatigue (since not all joints need to be viewed)
Repeat all joint ID's desired (total equals JOINTS).

The example below shows a typical fatigue extract summation file for three simulations with 497 joints chosen to view.

```

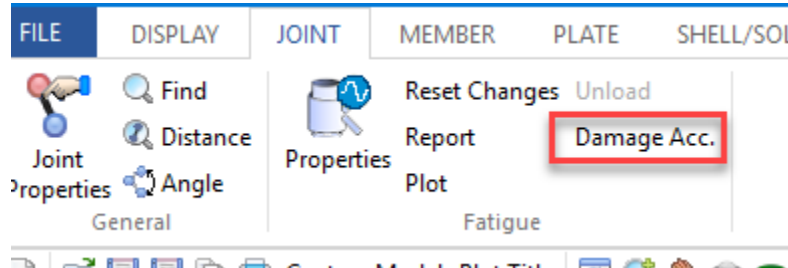
SUMFTGIFT
NCASES,3
NGRPS,2
DIRECTORY,"D:\Bentley\000Sample\1"
EXTRACT,"ftgext.dat"
GROUP,1
DIRECTORY,"D:\Bentley\000Sample\2"
EXTRACT,"ftgext.dat"
GROUP,2
DIRECTORY,"D:\Bentley\000Sample\3"
EXTRACT,"ftgext.dat"
GROUP,2
JOINTS,          497
  1
  3
  5
  7
 24
 71
 73
 75
 77

```

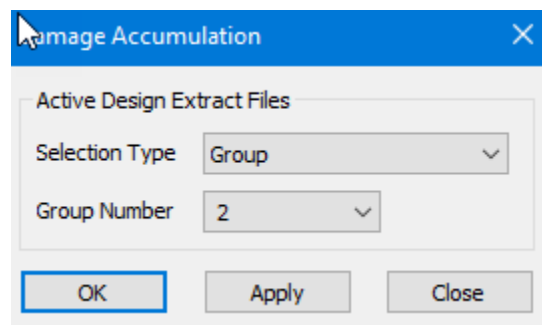
An extract summation file may be imported into Precede in the same manner as a normal fatigue extract file.

The initial state is to have all referenced simulations loaded. The shown results are the summation across all referenced extract files.

5.7.1 Select Extract File Or FileGroup



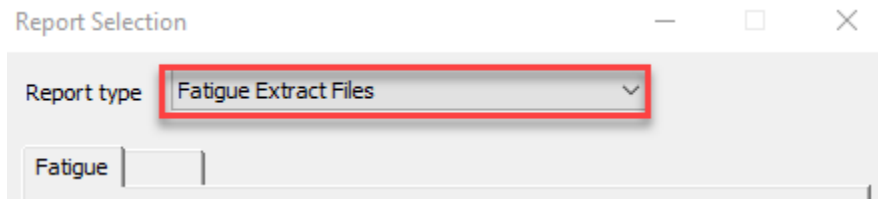
The Damage Acc. Button launches a dialog allowing the selection of an individual extract file or of a predefined group of extract files. When this is done only the selected extract file(s) is considered for redesign and summation of results. Note that redesign parameters are not reset between switching extract files.



Selection type may be “All Extract Files”, “Group” or “Single Extract File”. The lower box allows picking of the specific item.

5.7.2 DISPLAY EXTRACT FILES

A report of all extract files in the summation file is available from the report property box.



Contents brings up the list of extract file associated with this summation run.

